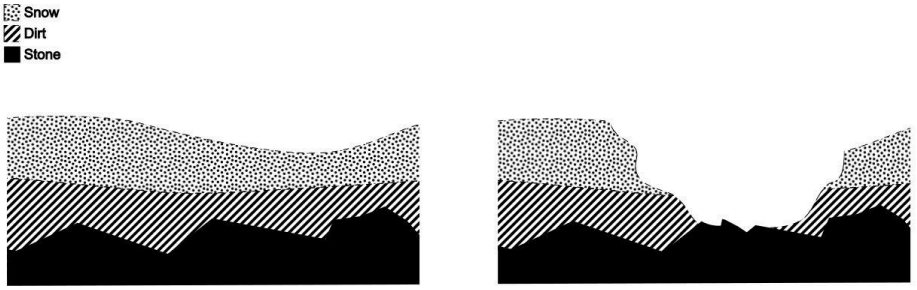
Objective:
This research aims to evaluate real-time deformation requests coming from the CPU, the main processor of the computer, and transfer them to the GPU and provide a believable simulation of a multi-layer material deformation. This should be done at reasonable, scalable GPU Memory costs, at a real-time frame rate, and in a way that integrates with existing game engines.

To illustrate how this would work, we present the following diagram.

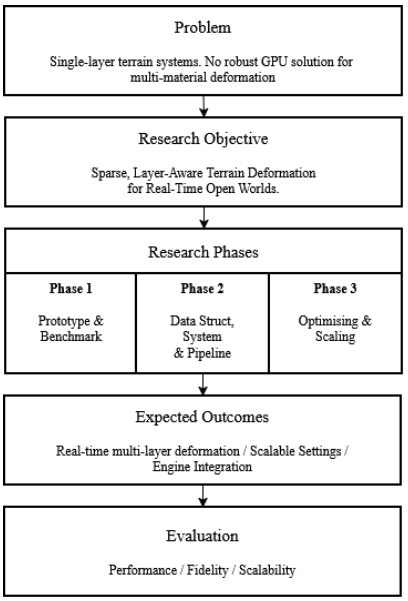


There are three materials stacked on top of each other: snow, dirt, and stone. A given deformation, with the same properties of strength, shape and size, is applied to the terrain. Because of the properties of different materials, the deformation would propagate less and less as it covers distance. Another result would be how the different layers are eroded: snow is fluffier, dirt is a bit denser, and stone is the densest. Therefore, Snow is scooped, but stone is chipped.

Method:
The research is structured in five stages: problem identification, objective setting, three development phases, expected outcomes and final evaluation.

The main research phases are as follows:

1. **Prototyping the system**: Answering the questions of how materials interact with each other, benchmarking different data storage methods and how the properties of the materials would affect the deformation.
2. **Deformation Data & System Structure**: Choosing a candidate for data storage and mapping how the system will be structured. This includes methods for transferring data from the CPU to the GPU and for making the final deformation available for sampling. Evaluate memory usage, fidelity and performance.
3. **Optimising & Scaling**: Using the previous metrics, develop optimisation methods. Scaling based on distance, caching unused deformation chunks… To improve real-time performance that can be set up easily by other developers on their projects.



Research Features:
This research differentiates itself in the overlap between Sparse GPU Data Structures, physically-based terrain deformation and real-time rendering. Usually, these areas are studied separately, rarely combined into a single real-time system targeting game engine integration. I aim to provide a strong solution that covers all these requirements while keeping it accessible to developers using existing engines like Unreal Engine. Leveraging the experience of the Mukai Laboratory on physically-based simulations and interactive graphics will be of great help to this research.